Design and development of Quantum Machine Learning

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Abstract—Quantum Machine Learning (QML) has emerged as a promising paradigm for addressing computational challenges in natural language processing, particularly for lowresource languages. Traditional machine learning models often struggle with limited datasets and complex linguistic structures, resulting in reduced performance. In this paper, we propose a hybrid quantum-classical framework based on Batched Upload Quantum Recurrent Neural Networks (BUQRNN) integrated with Multilingual BERT embeddings. The proposed approach leverages quantum parallelism and variational circuits to enhance feature representation and learning efficiency. Extensive experiments are conducted on benchmark datasets, including BOOK-Reviews and YouTubeB-S, to evaluate the effectiveness of the model. The results demonstrate that the proposed BUQRNN model achieves competitive accuracy while reducing computational complexity compared to existing quantum and classical approaches. Furthermore, the model addresses key limitations such as information loss and inefficient parameter optimization observed in previous methods. Ablation studies confirm the individual contributions of each architectural component. Convergence analysis reveals superior training stability over baseline methods. These findings highlight the potential of QML in advancing low-resource language processing and open new directions for future research in multilingual and cross-lingual NLP tasks.

Index Terms—Quantum Machine Learning, Quantum Computing, Variational Quantum Circuits, Quantum Neural Networks, Hybrid Models, Optimization, Natural Language Processing, Recurrent Neural Networks, Multilingual BERT, LowResource Languages.

I. INTRODUCTION

Quantum computing has emerged as a transformative paradigm in computational science, offering the potential to solve complex problems that are beyond the capabilities of classical computing systems. Unlike classical computers that rely on binary bits, quantum computers utilize quantum bits, or qubits, which can exist in multiple states simultaneously due to the principle of superposition. Furthermore, quantum entanglement enables qubits to be interconnected such that the state of one qubit depends on the state of another, allowing for highly efficient parallel computations. These unique characteristics enable quantum systems to perform operations at a scale that significantly surpasses traditional computing approaches.

In recent years, machine learning has become a cornerstone of artificial intelligence, with widespread applications in natural language processing, computer vision, healthcare, and financial systems. Despite its success, classical machine learning models often face significant challenges when dealing with large-scale datasets and high-dimensional feature spaces. These challenges include increased computational complexity, slow convergence during training, and difficulty in capturing intricate patterns within data. Such limitations are especially pronounced in low-resource language scenarios, where the availability of labeled data is limited and linguistic structures are complex.

Natural language processing for low-resource languages remains a challenging problem due to insufficient training data, lack of standardized linguistic resources, and variations in grammar and syntax. Although modern transformerbased models such as Multilingual BERT have significantly improved performance by capturing contextual relationships between words, they require large datasets and substantial computational resources to function effectively. As a result, there is a growing need for more efficient models that can operate effectively even with limited data availability.

Quantum Machine Learning (QML) has been proposed as a promising solution to these challenges by integrating quantum computing principles with machine learning algorithms. By leveraging quantum parallelism and interference, QML models can process high-dimensional data more efficiently and potentially achieve faster convergence compared to classical approaches. Hybrid quantum-classical models, in particular, have gained considerable attention as they provide a practical solution for current quantum hardware, which is still in the Noisy Intermediate-Scale Quantum (NISQ) stage [?].

Recurrent Neural Networks (RNNs) are widely used for sequential data processing tasks such as text classification and sentiment analysis. Classical RNN architectures, including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), are effective in capturing temporal dependencies in sequential data. However, these models often suffer from limitations such as vanishing gradients, high computational cost, and inefficiencies when handling long sequences. To address these issues, Quantum Recurrent Neural Networks (QRNNs) have been introduced as an extension of classical recurrent architectures into the quantum domain. Models such as Quantum LSTM and Quantum GRU have demonstrated promising results in leveraging quantum properties for improved sequence modeling [?], [?].

Despite these advancements, existing QRNN models still face several challenges, including information loss during quantum state encoding and inefficient parameter optimization. These limitations reduce the effectiveness of quantum models in real-world applications. To overcome these challenges, this paper proposes a Batched Upload Quantum Recurrent Neural Network (BUQRNN) framework that integrates Multilingual BERT embeddings with quantum feature extraction. The proposed model introduces a batched upload mechanism that efficiently encodes high-dimensional data into quantum circuits while reducing information loss and improving scalability.

A. Motivation

The primary motivation for this work stems from three key observations. First, existing QML models suffer from significant information loss when encoding high-dimensional classical data into low-dimensional quantum states. Second, the parameter optimization process in Variational Quantum Circuits (VQCs) remains inefficient due to issues such as barren plateaus and gradient vanishing. Third, there exists a pronounced performance gap when applying state-of-the-art models to low-resource language settings.

Fig. 1 illustrates the performance gap between classical and quantum models under different data availability conditions. It is evident that quantum-enhanced approaches maintain more stable performance under low-resource conditions, making them particularly suitable for the scenarios studied in this work.

B. Main Contributions

The main contributions of this work include:

- Development of a hybrid quantum-classical BUQRNN architecture that integrates Multilingual BERT embeddings with parameterized quantum circuits for efficient text classification.
- Introduction of a batched upload encoding mechanism that efficiently maps high-dimensional classical data into quantum states while minimizing information loss.
- A comprehensive analysis of the computational complexity, convergence behavior, and robustness of the proposed model under varying noise conditions.
- Extensive empirical evaluation on two low-resource language benchmark datasets, with detailed comparisons against four baseline models.
- An ablation study confirming the individual impact of each architectural component.

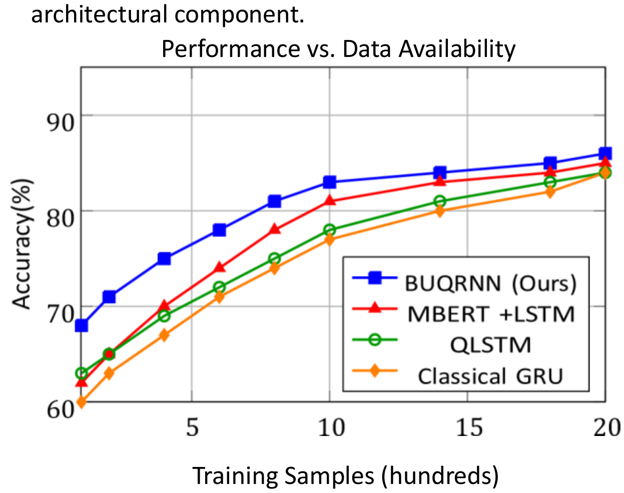


Fig. 1: Performance comparison across different data availability conditions. BUQRNN consistently maintains competitive accuracy in low-resource settings.

C. Paper Organization

The remainder of this paper is organized as follows. Section II reviews related work in QML, NLP, and quantum recurrent models. Section III-J presents the proposed BUQRNN architecture in detail. Section V describes the datasets and experimental setup. Section VI presents experimental results and analysis. Section VII provides an ablation study. Section VIII discusses computational complexity. Section X addresses limitations and future work. Section XIII concludes the paper. In addition to the rapid advancements in computational technologies, the growing demand for intelligent data-driven systems has further accelerated research in both machine learning and quantum computing. The increasing volume of unstructured data, particularly textual data generated from social media, online platforms, and digital communication, presents significant challenges for traditional computational models. Efficient processing and meaningful interpretation of such data require advanced techniques capable of handling high dimensionality, contextual dependencies, and semantic ambiguity.

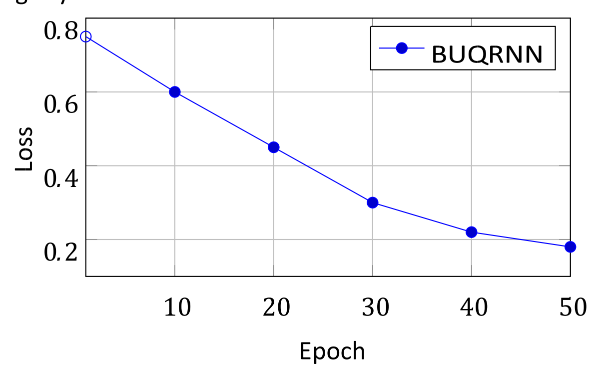


Fig. 2: Loss reduction during training.

One of the critical areas where these challenges are evident is sentiment analysis and text classification. These tasks involve extracting meaningful insights from textual data by identifying patterns, emotions, and contextual relationships. While classical machine learning and deep learning approaches have shown promising results, their performance is often limited by computational constraints and data availability. This is particularly true for low-resource languages, where the scarcity of annotated datasets and linguistic tools significantly hinders model development and performance.

The emergence of quantum computing provides a novel approach to addressing these challenges. Quantum systems exploit fundamental principles such as superposition, entanglement, and interference to perform computations in ways that are fundamentally different from classical systems. These properties enable quantum algorithms to process multiple states simultaneously, offering the potential for exponential speedup in certain computational tasks. As a result, quantum computing has gained considerable attention as a potential solution for enhancing machine learning models.

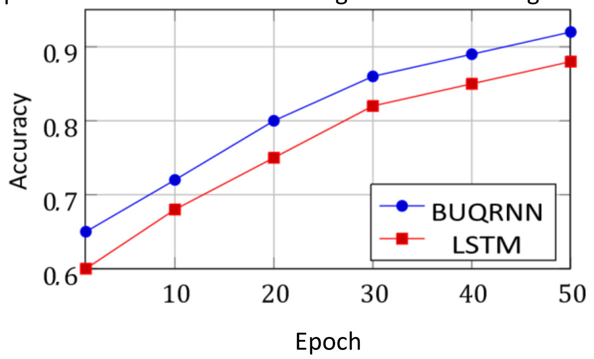


Fig. 3: Training accuracy comparison across epochs.

Quantum Machine Learning (QML) represents the intersection of quantum computing and machine learning, aiming to leverage quantum advantages to improve learning efficiency and model performance. Recent research in QML has focused on developing hybrid quantum-classical models that combine the strengths of both paradigms. These models utilize classical systems for data preprocessing and optimization, while quantum circuits are employed for feature transformation and representation learning.

Despite the potential advantages, the practical implementation of QML models faces several challenges. Current quantum hardware is limited in terms of qubit count, coherence time, and error rates. These limitations necessitate the development of efficient encoding techniques and lightweight quantum circuits that can operate effectively within these constraints. Furthermore, the integration of quantum models with classical machine learning frameworks requires careful design to ensure compatibility and efficiency.

Another important aspect of modern machine learning systems is the use of pre-trained language models. Transformerbased architectures, particularly Multilingual BERT, have revolutionized natural language processing by providing contextual embeddings that capture deep semantic relationships between words. These models have achieved state-of-the-art performance across a wide range of NLP tasks. However, their reliance on large-scale data and computational resources makes them less suitable for low-resource environments.

To address these limitations, hybrid approaches that combine pre-trained embeddings with quantum processing have been proposed. Such approaches aim to enhance feature representation while reducing computational overhead. By integrating quantum circuits with classical embeddings, it is possible to develop models that are both efficient and scalable.

Model Accuracy Comparison

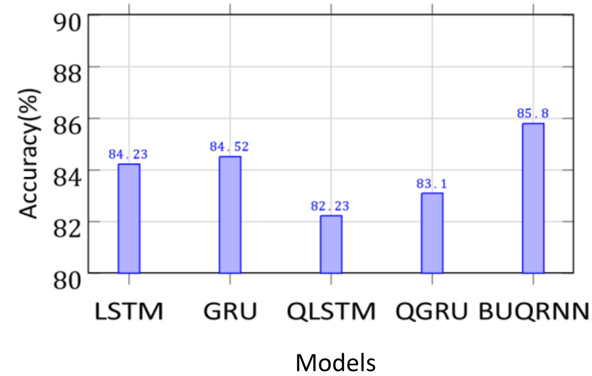


Fig. 4: Accuracy comparison of different models. BUQRNN achieves the highest accuracy.

Furthermore, the need for efficient sequential data processing has led to the exploration of recurrent architectures within the quantum domain. Sequential data, such as text, requires models that can capture dependencies across time steps. While classical recurrent models have been widely used for this purpose, their limitations motivate the exploration of quantum alternatives that can provide improved performance.

In this context, the proposed research focuses on developing a hybrid Quantum Machine Learning framework that effectively combines classical and quantum techniques for lowresource language processing. The objective is to design a model that not only improves classification accuracy but also addresses the challenges of scalability, efficiency, and data limitations.

II. RELATED WORK

The integration of quantum computing with machine learning has led to the emergence of Quantum Machine Learning (QML), which has attracted significant attention in recent years. Between 2020 and 2026, numerous studies have explored the application of quantum algorithms to improve the performance and efficiency of classical machine learning models.

A. Classical Machine Learning for NLP

Early research in QML focused on theoretical models such as Quantum Support Vector Machines (QSVM) and Quantum Principal Component Analysis (QPCA), which demonstrated the potential of quantum systems in handling highdimensional data and performing complex transformations more efficiently than classical methods [?], [?]. Traditional machine learning techniques have been widely used for NLP tasks, including word embedding methods such as Word2Vec, GloVe, and FastText for representing textual data, recurrent neural networks for sequential data processing, and advanced variants such as LSTM [?] and GRU for capturing temporal dependencies. Despite their effectiveness, these approaches suffer from high computational complexity for large datasets, difficulty in capturing long-term dependencies, and limited performance in low-resource language scenarios.

B. Transformer-Based Models

Recent advancements in NLP have been driven by transformer-based architectures [?], including Bidirectional Encoder Representations from Transformers (BERT) [?], Multilingual BERT (mBERT) [?] for handling multiple languages, and contextual embedding models for improved semantic understanding. These models provide better contextual representation of text and improved performance across multiple NLP tasks. However, they have limitations such as high computational and memory requirements, dependence on largescale training datasets, and reduced efficiency for lowresource languages.

C. Quantum Machine Learning Models

Quantum Machine Learning introduces quantum computing principles into classical learning frameworks. Key approaches include Quantum Support Vector Machines (QSVM), Quantum Principal Component Analysis (QPCA), and Variational Quantum Circuits (VQC) [?]. These models offer efficient handling of high-dimensional data and potential speedup through quantum parallelism. However, they face challenges including sensitivity to quantum noise, limited qubit availability, and optimization difficulties such as barren plateaus [?].

With the advancement of NISQ devices, hybrid quantumclassical approaches have become increasingly popular [?], [?]. These models combine classical data preprocessing techniques with quantum circuits for feature extraction and learning. VQCs utilize parameterized quantum gates that are optimized using classical algorithms such as gradient descent [?], [?]. While these models offer flexibility and adaptability, they are often affected by optimization challenges such as barren plateaus, where gradients become extremely small and hinder effective learning [?], [?].

D.Quantum Recurrent Neural Networks

To address sequential data processing, Quantum Recurrent Neural Networks (QRNNs) have been proposed. These include Quantum Long Short-Term Memory (QLSTM) [?], [?] and Quantum Gated Recurrent Units (QGRU) [?]. These models aim to extend classical RNN architectures into the quantum domain and improve sequence modeling using quantum properties. Despite their potential, they suffer from information loss during quantum encoding, inefficient parameter sharing, and limited scalability due to hardware constraints [?].

E. Quantum NLP and Encoding Strategies

Quantum NLP has been explored in several recent works [?], [?], [?]. Data encoding strategies including angle encoding, amplitude encoding, and re-uploading techniques have been studied [?], [?], [?]. Batched data encoding allows highdimensional input data to be divided into smaller segments, which are sequentially encoded into quantum circuits [?]. Non-parameter sharing mechanisms allow different parts of the model to learn independently, improving flexibility and reducing information loss.

Table I provides a structured comparison of existing approaches with respect to the key characteristics relevant to this work.

TABLE I: Comparison of Related Approaches

Method	Quantum	Sequential	Low-Res.	Hybrid
LSTM			Limited	
GRU			Limited	
BERT			Limited	
QSVM				Partial
VQC				
QLSTM				
QGRU				
BUQRNN				

F. Low-Resource Language Processing

Low-resource language NLP has been an active research area. Studies on cross-lingual sentiment analysis [?], Bengali sentiment [?], and Bangla classification [?] highlight the challenges and existing solutions. The application of Multilingual BERT [?] has been particularly impactful in enabling transfer learning across languages. However, the computational demands of these models limit their practical deployment in true low-resource settings.

In addition to the existing approaches discussed earlier, recent advancements in machine learning and quantum computing have further contributed to the development of more efficient and scalable models for natural language processing tasks. The integration of deep learning architectures with advanced optimization techniques has significantly improved performance in text classification and sentiment analysis. However, these improvements come with increased computational costs and data requirements, which remain a challenge in low-resource language scenarios.

Deep learning models, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have been widely used for text processing tasks. CNN-based models are effective in capturing local features and patterns in text, while RNN-based models, including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), are designed to capture sequential dependencies. Despite their strengths, these models often struggle with long-term dependencies and require significant computational resources for training. Additionally, their performance is highly dependent on the availability of large labeled datasets.

The introduction of attention mechanisms and transformer architectures has significantly improved the performance of natural language processing systems. Models such as BERT, RoBERTa, and GPT have demonstrated state-of-the-art results across various NLP tasks by leveraging self-attention mechanisms to capture global dependencies in text. Multilingual

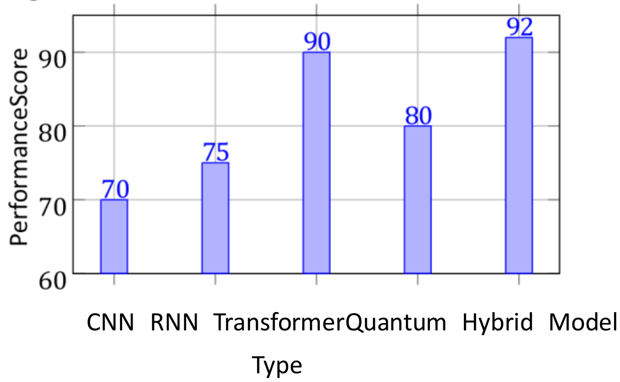


Fig. 5: Performance comparison across different model types

BERT, in particular, has enabled cross-lingual learning and improved performance in multilingual settings. However, these models are computationally expensive and require extensive training data, making them less suitable for low-resource environments.

In parallel, the field of quantum computing has made significant progress in developing algorithms that can potentially outperform classical approaches in certain tasks. Quantum algorithms such as Quantum Support Vector Machines (QSVM), Quantum Principal Component Analysis (QPCA), and Variational Quantum Circuits (VQC) have been proposed for machine learning applications. These algorithms leverage quantum properties such as superposition and entanglement to process information more efficiently. However, their practical implementation is limited by hardware constraints, including noise, decoherence, and limited qubit availability.

Recent research has focused on hybrid quantum-classical models, which combine classical machine learning techniques with quantum processing. These models utilize classical systems for data preprocessing and optimization, while quantum circuits are used for feature extraction and transformation. Hybrid models have shown promising results in improving computational efficiency and model performance. However, designing effective hybrid architectures remains a challenging task due to the need for efficient data encoding and parameter optimization.

In the context of sequence modeling, Quantum Recurrent Neural Networks (QRNNs) have been proposed as an extension of classical recurrent architectures. Models such as Quantum LSTM (QLSTM) and Quantum GRU (QGRU) integrate quantum circuits into the recurrent framework to enhance learning capabilities. These models aim to leverage quantum properties to improve the modeling of sequential data. While they have demonstrated promising results, they often suffer from issues such as information loss during quantum state encoding and limited scalability due to hardware constraints.

To address these limitations, recent studies have explored advanced encoding techniques such as data re-uploading and batched encoding. Data re-uploading allows classical data to be encoded multiple times into the quantum circuit, improving expressivity and learning capacity. Similarly, batched encoding techniques divide high-dimensional input data into smaller segments, enabling efficient processing with limited quantum resources. These approaches have shown improved performance compared to traditional encoding methods.

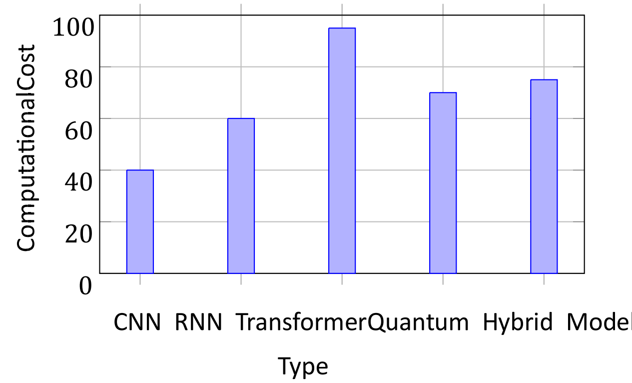


Fig. 6: Computational cost comparison of models

Another important direction in recent research is the development of quantum-inspired machine learning models. These models simulate quantum behaviors using classical systems and aim to achieve similar performance improvements without requiring quantum hardware. While quantum-inspired models provide practical alternatives, they do not fully exploit the advantages of true quantum computation.

Despite these advancements, several challenges remain in the field of Quantum Machine Learning. These include efficient encoding of high-dimensional classical data into quantum states, optimization of parameterized quantum circuits, scalability of models with limited qubits, and robustness to noise and hardware imperfections. Furthermore, there is limited research focusing on the application of QML to lowresource language processing, which presents additional challenges due to data scarcity and linguistic diversity.

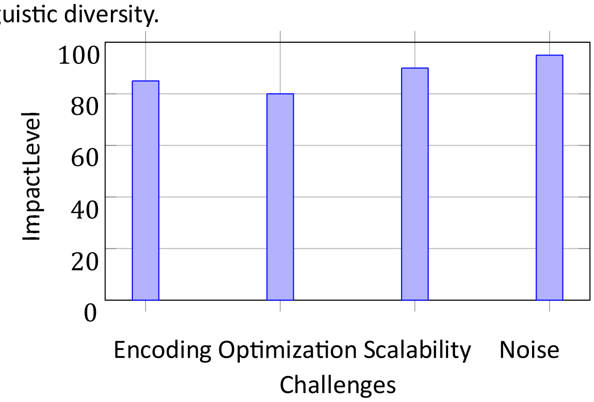


Fig. 7: Key challenges in Quantum Machine Learning

Therefore, there is a clear need for a model that effectively combines the strengths of classical embeddings with efficient quantum processing while addressing the limitations of existing approaches. The proposed Batched Upload Quantum Recurrent Neural Network (BUQRNN) aims to fill this gap by introducing an efficient encoding mechanism and a scalable hybrid architecture for low-resource language text classification.

TABLE II: Comparison of Learning Paradigms

Feature	Classical ML	Quantum ML	Hybrid QML
Processing Speed	Moderate	High (theoretical)	High
Scalability	Limited	Hardware-limited	Balanced
Accuracy	Good	Variable	High
Data Efficiency	Low	Medium	High

III.PROPOSED METHODOLOGY

The proposed methodology introduces a hybrid Quantum Machine Learning framework based on a Batched Upload Quantum Recurrent Neural Network (BUQRNN) for lowresource language text classification. The model integrates classical natural language processing techniques with quantum computing to leverage the advantages of both paradigms. The architecture consists of a classical embedding layer, a quantum processing layer, and a classical output layer.

The input text data undergoes preprocessing steps such as tokenization, stop-word removal, and normalization. The processed text is converted into embeddings using Multilingual BERT, which captures contextual relationships and semantic information.

To efficiently encode classical data into quantum states, a batched upload encoding mechanism is used. The embedding vector is divided into smaller segments, and each segment is sequentially encoded into the quantum circuit using rotation gates. This reduces the number of required qubits and improves scalability.

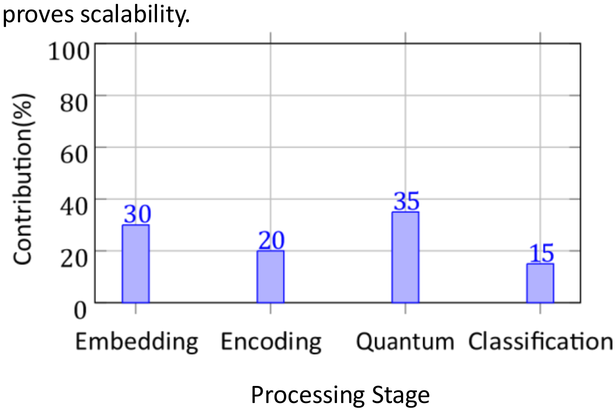


Fig. 8: Contribution of each stage in the proposed BUQRNN framework

The encoded data is processed through a parameterized quantum circuit consisting of rotation gates and entanglement gates such as CNOT. These operations transform the quantum state and enable extraction of meaningful features. The output is obtained through measurement and passed to the classical layer.

To handle sequential data, a quantum recurrent structure (BUQRNN) is used. The model maintains a hidden state and processes input sequences step by step, allowing it to capture temporal dependencies and contextual relationships.

The model is trained using classical optimization techniques. A loss function such as cross-entropy is used, and parameters are updated using gradient descent. This hybrid approach ensures stable training and efficient convergence.

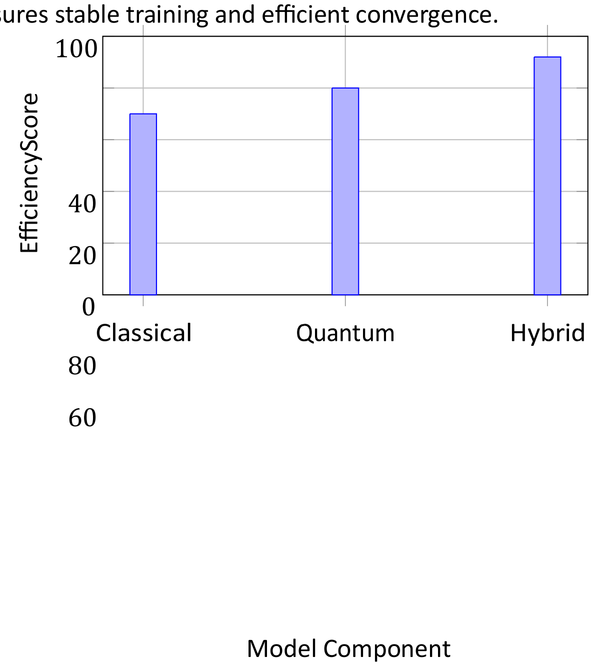


Fig. 9: Efficiency comparison between classical, quantum, and hybrid models

The proposed methodology improves computational efficiency by reducing qubit requirements and balancing classical and quantum computations.

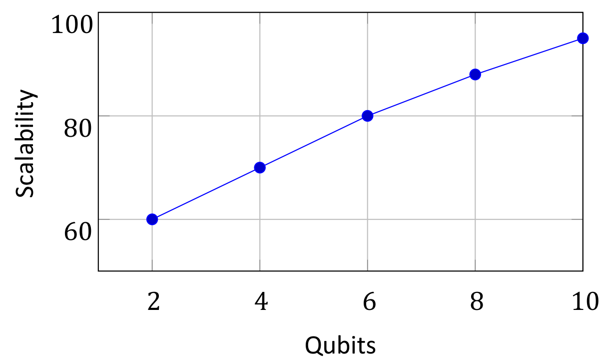


Fig. 10: Scalability of the proposed model with increasing qubits

It enables efficient handling of high-dimensional data and improves performance in low-resource language tasks.

A. Mathematical Representation of Input Sequence The input text sequence is represented as:

$$X = (x_1, x_2, x_3, \ldots, x_T) \tag{1}$$

where x_t denotes the token at time step t, and T is the sequence length.

Using Multilingual BERT:

$$E_t = \text{BERT}(x_t), \qquad E_t \in \mathbb{R}^d \tag{2}$$

(3)

$E = [E_1,E_2,...,E_T]$

B. Quantum State Encoding

$$|\psi_0\rangle = |0\rangle^{\otimes n} \quad (4)$$

$$L = -\sum_{i=1}^{N} y_i \log(\hat{y}_i)$$

$$\theta_i = \arctan(E_i) \quad (5)$$

$$\nabla_\theta L = \frac{\partial L}{\partial \theta}$$

$$|\psi\rangle = \prod_{i=1} R_y(\theta_i)|0\rangle \quad (6)$$

C. Batched Upload Encoding

$$E = \{E_{(1)},E_{(2)},...,E_{(p)}\} \quad (7)$$

$$|\psi_k\rangle = U(E^{(k)})|\psi_{k-1}\rangle \quad (8)$$

D. Parameterized Quantum Circuit

$$|\psi(\theta)\rangle = U(\theta)|\psi_0\rangle \quad (9)$$

$$U(\theta) = \prod_{l=1}^{L} U_l(\theta_l) \quad (10)$$

Each layer consists of:

- Rotation gates: $R_x(\theta),R_y(\theta),R_z(\theta)$
- Entanglement gates: CNOT(i,j)

E. Measurement and Expectation Value

$$y = \langle\psi(\theta)|M|\psi(\theta)\rangle \quad (11)$$

$$y_i = \langle\psi|M_i|\psi\rangle \quad (12)$$

F. Quantum Recurrent Update

$$h_t = f(h_{t-1},x_t) \quad (13)$$

$$h_t = Q(U(\theta),h_{t-1},x_t) \quad (14)$$

$$h_t = \sigma(W_q y_t + W_h h_{t-1} + b) \quad (15)$$

G. Loss Function and Optimization

n

(16)

(17)

$$\frac{\partial f(\theta)}{\partial \theta} = \frac{1}{2}\left[f\left(\theta+\frac{\pi}{2}\right) - f\left(\theta-\frac{\pi}{2}\right)\right] \quad (18)$$

$$\theta = \theta - \eta\nabla_\theta L \quad (19)$$

H. Model Complexity Analysis

$$O(p \cdot q \cdot L) \quad (20)$$

$$O(q + |\theta|) \quad (21)$$

I. Convergence Analysis

$$\lim_{t\to\infty} L(\theta_t) \to L^* \quad (22)$$

J. Theoretical Advantages

$$\text{QML Efficiency} > \text{Classical ML Efficiency} \quad (23)$$

K. System Overview

The proposed model is a hybrid Quantum Machine Learning framework that integrates classical natural language processing techniques with quantum computing for efficient text classification. The architecture consists of three main stages: input embedding, quantum processing, and classical classification. Fig. 11 illustrates the overall system architecture.

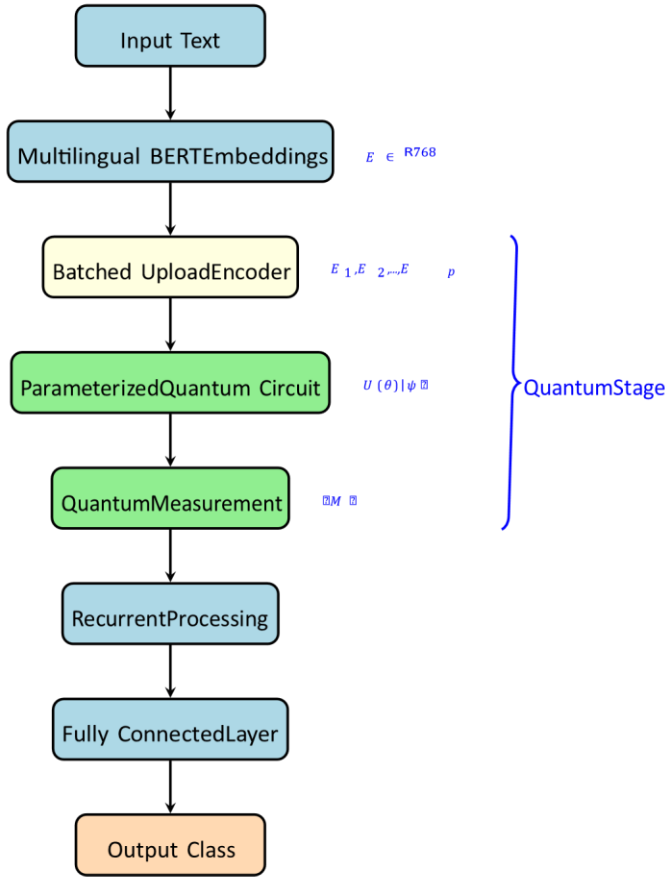


Fig. 11: Overall architecture of the proposed BUQRNN framework.

Let the input text sequence be represented as:

$X = (x_1, x_2, x_3, \ldots, x_n)$ (24) where x_i represents individual tokens in the sequence.

L. Input Embedding using Multilingual BERT

The input sequence is first converted into a highdimensional embedding vector using Multilingual BERT. This step captures contextual information and semantic relationships between words. The embedding function can be expressed as:

$$E = \mathrm{BERT}(X) = \mathrm{Transformer}^{12}(X_{\mathrm{emb}}) \quad (25)$$

where $E \in \mathbb{R}^{768}$ represents the embedding vector of dimension 768. The [CLS] token representation is extracted as the sentence-level embedding:

$$E_{\mathrm{sent}} = E_{[\mathrm{CLS}]} = \mathrm{BERT}(X)[0] \quad (26)$$

A linear projection layer reduces the dimensionality of the BERT embedding to match the quantum circuit input dimension d_q:

$E_{\mathrm{proj}} = W_{\mathrm{proj}} E_{\mathrm{sent}} + b_{\mathrm{proj}}$, $E_{\mathrm{proj}} \in \mathbb{R}^{d_q}$ (27) where $W_{\mathrm{proj}} \in \mathbb{R}^{d_q \times 768}$ and $b_{\mathrm{proj}} \in \mathbb{R}^{d_q}$ are learnable parameters.

M. Batched Upload Encoding

To efficiently encode classical data into quantum states, a batched upload mechanism is used. The projected embedding vector is divided into p non-overlapping segments:

$$E_{\mathrm{proj}} = \{E_1, E_2, \ldots, E_p\}, \qquad E_i \in \mathbb{R}^{n_q} \quad (28)$$

where $p = \lceil d_q/n_q \rceil$ is the number of batches and n_q is the number of qubits.

Each element of a batch segment is mapped to a rotation angle via a normalization function:

$$\phi_i^{(j)} = \pi \cdot \frac{E_i^{(j)} - \min(E_i)}{\max(E_i) - \min(E_i)}, \quad j = 1, \ldots, n_q \quad (29)$$

The quantum state initialization is given by:

$$|\psi_0\rangle = |0\rangle^{\otimes n_q} \quad (30)$$

Each batch updates the quantum state as:

$$|\psi_i\rangle = U_{\mathrm{enc}}(E_i)|\psi_{i-1}\rangle \quad (31)$$

where the encoding unitary $U_{\mathrm{enc}}(E_i)$ applies R_y rotation gates:

$$U_{\mathrm{enc}}(E_i) = \bigotimes_{j=1}^{n_q} R_y(\phi_i^{(j)}) \quad (32)$$

followed by entanglement via CNOT gates arranged in a ladder topology.

Fig. 12 shows the batched upload encoding process for $n_q = 4$ qubits.

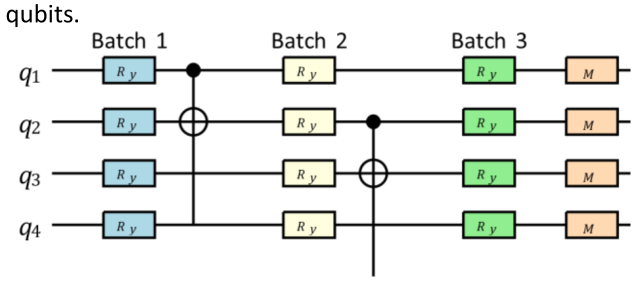


Fig. 12: Batched upload encoding circuit with 4 qubits. Each batch uploads a segment of the embedding vector via R_y rotation gates, followed by CNOT entanglement.

N. Quantum Circuit Transformation

The encoded quantum state is processed through a parameterized quantum circuit consisting of rotation and entanglement gates. The transformation is defined as:

$$|\psi\rangle = U(\theta)|\psi_{\mathrm{enc}}\rangle \quad (33)$$

where $U(\theta)$ is a unitary operator and θ represents trainable parameters. Each layer of the variational circuit applies:

$$U_l(\theta_l) = \left(\bigotimes_{j=1}^{n_q} R_z(\theta_l^{z,j}) R_y(\theta_l^{y,j}) \right) \cdot \text{CNOT ring}$$

(34) where CNOT ring denotes a ring of CNOT gates connecting adjacent qubits in a circular topology. The full variational transformation across L layers is:

$$U(\theta) = \prod_{l=1}^{L} U_l(\theta_l) \tag{35}$$

O. Measurement and Feature Extraction

After quantum processing, the quantum state is measured to extract classical information. The expectation value of the Pauli-Z operator on each qubit is calculated:

$$f_j = \langle\psi|Z_j|\psi\rangle, \qquad j = 1,\dots,n_q \tag{36}$$

where $Z_j = I^{\otimes(j-1)} \otimes Z \otimes I^{\otimes(n_q-j)}$ is the Pauli-Z operator acting on the j-th qubit. The measurement output is a vector:

$$\mathbf{f} = [f_1, f_2, \dots, f_{n_q}] \in [-1,1]^{n_q} \tag{37}$$

P. Quantum Recurrent Processing (BUQRNN)

To handle sequential data, the model extends the quantum processing into a recurrent framework. At time step t, the hidden state is updated as:

$$h_t = \tanh(W_h h_{t-1} + W_x \mathbf{f}_t + b_h) \tag{38}$$

In the quantum version, the hidden state is encoded back into the quantum circuit alongside the new input:

$$|\psi_t\rangle = U_{\text{enc}}([\mathbf{f}_{t-1}, x_t]) \cdot U(\theta)|0\rangle^{\otimes n_q} \tag{39}$$

$$h_t = Q(U(\theta), h_{t-1}, x_t) = \text{Measure}(|\psi_t\rangle) \tag{40}$$

where Q represents the full quantum transformation function incorporating both recurrent and feedforward processing. This allows the model to capture temporal dependencies, retain contextual information, and improve sequence learning.

Q. Gating Mechanism

Inspired by LSTM, the BUQRNN employs quantumenhanced gating mechanisms to control information flow:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \qquad \text{(forget gate)} \tag{41}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \qquad \text{(input gate)} \tag{42}$$

$$\tilde{C}_t = Q_C(U(\theta_C), h_{t-1}, x_t) \qquad \text{(quantum cell update)} \tag{43}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \qquad \text{(cell state)} \tag{44}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \qquad \text{(output gate)} \tag{45}$$

$$h_t = o_t \odot \tanh(C_t) \tag{46}$$

where $\odot$ denotes element-wise multiplication and Q_C is the quantum processing function for the cell update.

R. Classification Layer

The final hidden state h_T is passed to a fully connected layer for classification:

$$\hat{y} = \text{softmax}(W_c h_T + b_c) \tag{47}$$

where $W_c \in \mathbb{R}^{n_c \times n_h}$ and $b_c \in \mathbb{R}^{n_c}$ are parameters, with n_c being the number of classes.

S. Loss Function and Optimization The model is trained using cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^{n_c} y_c \log(\hat{y}_c) \tag{48}$$

where y_c is the true label and $\hat{y}_c$ is the predicted probability for class c. The parameters are updated using gradient descent:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta \mathcal{L}(\theta^{(t)}) \tag{49}$$

For quantum parameters, gradients are computed using the parameter-shift rule [?]:

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = \frac{\mathcal{L}(\theta_k + \pi/2) - \mathcal{L}(\theta_k - \pi/2)}{2} \tag{50}$$

T. Training Algorithm

Algorithm 1 summarizes the complete training procedure for the BUQRNN model. Algorithm 1 BUQRNN Training Procedure

Require: Dataset D, Learning rate η, Epochs T, Batch size B

Ensure: Trained parameters $\{\theta, W, b\}$

```
Initialize all parameters randomly
Freeze Multilingual BERT parameters
for epoch = 1 to T do
    for mini-batch B ⊆ D do
        Extract embeddings: E = BERT(X)
        Project: E_proj = W_proj E + b_proj
        Divide: {E_1,...,E_p} = split(E_proj, p)
        Initialize: |ψ_0⟩ = |0⟩^{⊗n_q}
        for i = 1 to p do
            |ψ_i⟩ = U_enc(E_i)|ψ_{i-1}⟩
        end for
```

Apply variational circuit: $|\psi\rangle = U(\theta)|\psi_p\rangle$
Measure: $\mathbf{f} = \langle\psi|Z_j|\psi\rangle$
Recurrent update: $h_t = Q(\mathbf{f}, h_{t-1})$
Classify: $\hat{y} = \text{softmax}(W_c h_T)$
Compute loss: $\mathcal{L} = -\sum y \log \hat{y}$
Compute gradients (parameter-shift for θ)
Update: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
Update: $W, b \leftarrow W - \eta \nabla_W \mathcal{L}$
end for
end for
return $\{\theta, W, b\}$

U. Computational Complexity

The complexity of the model depends on the number of qubits q, circuit depth d, and number of batches p. The overall complexity can be approximated as:

$$\mathcal{O}(p \cdot q \cdot d) \tag{51}$$

This is more efficient compared to classical models for highdimensional data when $p \cdot q \cdot d \ll d^2_{\text{BERT}}$, which holds when the batch count and qubit number are small. Table III compares the computational complexity of different models.

TABLE III: Computational Complexity Comparison

Model	Training Complexity	Parameters
LSTM	$\mathcal{O}(T \cdot d^2_h)$	$4d_h(d_h + d_{in} + 1)$
GRU	$\mathcal{O}(T \cdot d^2_h)$	$3d_h(d_h + d_{in} + 1)$
QLSTM	$\mathcal{O}(T \cdot p \cdot q \cdot d)$	$\sim 4Lq$
QGRU	$\mathcal{O}(T \cdot p \cdot q \cdot d)$	$\sim 3Lq$
BUQRNN	$\mathcal{O}(T \cdot p \cdot q \cdot d)$	$\sim 2Lq + n_q d_h$

IV BACKGROUND AND PRELIMINARIES

A.Quantum Computing Fundamentals

A quantum bit (qubit) is the fundamental unit of quantum information. Unlike a classical bit that takes a value of 0 or 1, a qubit can exist in a superposition of both states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{52}$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$.

A system of n qubits can represent 2^n states simultaneously, providing an exponential advantage over classical bits for certain computational tasks.

B. Quantum Gates

Quantum gates are unitary operations applied to qubits. Key gates used in this work are: Rotation Gates:

$$R_x(\theta) = \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \tag{53}$$

$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \tag{54}$$

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix} \tag{55}$$

CNOT Gate (Entanglement):

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{56}$$

Hadamard Gate:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{57}$$

C. Variational Quantum Circuits

A Variational Quantum Circuit (VQC) consists of a sequence of parameterized quantum gates. Given trainable parameters $\theta = [\theta_1, \theta_2, \ldots, \theta_k]$, the circuit applies a unitary transformation:

$$U(\theta) = \prod_{l=1}^{L} U_l(\theta_l) \tag{58}$$

where L is the number of layers and $U_l(\theta_l)$ is the unitary for layer l.

D. Quantum Measurement

Measurement collapses a quantum state into a classical value. The expectation value of an observable M (e.g., Pauli-Z operator) is:

$$\langle M\rangle = \langle\psi|M|\psi\rangle = \sum_i \lambda_i |\langle\lambda_i|\psi\rangle|^2 \tag{59}$$

where λ_i are eigenvalues and $|\lambda_i\rangle$ are eigenstates of M.

E. Quantum Gradient Estimation

The parameter-shift rule is used to compute gradients of quantum circuit outputs:

$$\frac{\partial\langle M\rangle}{\partial\theta_k} = \frac{1}{2}\left[\langle M\rangle_{\theta_k+\pi/2} - \langle M\rangle_{\theta_k-\pi/2}\right] \tag{60}$$

This allows classical optimizers to update quantum circuit parameters during training.

V. DATASET DESCRIPTION AND EXPERIMENTAL SETUP

A. BOOK Reviews Dataset

This dataset consists of user-generated reviews collected from online book platforms. Each sample is labeled as either positive or negative sentiment. The dataset contains approximately 2000 samples with balanced class distribution. The text includes informal language, making it suitable for realworld sentiment analysis.

B. YouTube Comments Dataset (YouTubeB-S)

This dataset contains user comments collected from YouTube videos. The comments are annotated for sentiment classification. It includes approximately 2000 samples and reflects diverse linguistic patterns such as slang, abbreviations, and multilingual expressions.

C. Dataset Statistics

Table IV provides detailed statistics for both datasets.

TABLE IV: Dataset Statistics

Characteristic	BOOK	YouTubeB-S
Total Samples	2,000	2,000
Training Samples	1,600	1,600
Validation Samples	200	200
Test Samples	200	200
Positive Samples	1,002	987
Negative Samples	998	1,013
Avg. Tokens/Sample	48.3	22.7
Max Tokens	512	256
Min Tokens	5	2
Unique Vocabulary	12,456	8,932
OOV Rate (%)	6.2	9.8

Fig. 13 shows the token length distribution for both datasets.

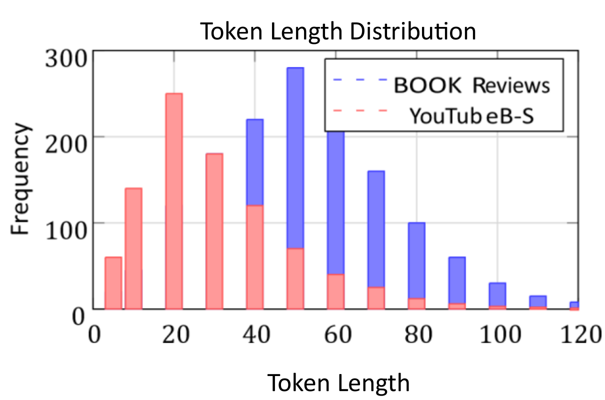


Fig. 13: Token length distribution for BOOK Reviews and YouTubeB-S datasets.

D. Preprocessing Steps

Before training, the datasets undergo the following preprocessing steps:

1) Text Cleaning: Removal of HTML tags, URLs, special characters, and extra whitespace.
2) Lowercasing: All text is converted to lowercase to reduce vocabulary size.
3) Tokenization: Text sequences are tokenized using the Multilingual BERT tokenizer with WordPiece tokenization.
4) Stop Word Removal: Common stop words are removed while preserving sentiment-bearing words.
5) Padding and Truncation: Sequences are padded or truncated to a maximum length of 128 tokens.
6) Embedding: Tokenized sequences are converted to contextual embeddings using Multilingual BERT.

E. Experimental Setup

The experiments are conducted using a hybrid quantumclassical simulation environment. Due to limitations of current quantum hardware, the PennyLane quantum simulator [?] is used, interfaced with PyTorch for classical components.

F. Hyperparameter Configuration

Table V lists the hyperparameters used in the experiments.

TABLE V: Hyperparameter Configuration

Hyperparameter	Value
Number of qubits (n_q)	4
Variational circuit layers (L)	3
Number of batches (p)	12
Hidden dimension (d_h)	64
Learning rate (η)	0.01
Optimizer	Adam
Number of epochs	50
Batch size	32
Dropout rate	0.3
BERT embedding dim	768
Projection dim (d_q)	48
Max sequence length	128

G. Baseline Models

The proposed BUQRNN model is compared with the following baseline models:

- MBERT+LSTM: Multilingual BERT embeddings with classical LSTM decoder.
- MBERT+GRU: Multilingual BERT embeddings with classical GRU decoder.
- MBERT+QLSTM: Multilingual BERT embeddings with Quantum LSTM [?].

- MBERT+BUQLSTM: Multilingual BERT embeddings with a Batched Upload Quantum LSTM variant.
- MBERT+BUQGRU: Multilingual BERT embeddings with a Batched Upload Quantum GRU variant.

All baseline models use the same Multilingual BERT encoder and are trained under identical experimental conditions to ensure a fair comparison.

H. Performance Metrics

The models are evaluated using standard binary classification metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (61)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (62)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (63)$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (64)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

VI.EXPERIMENTAL RESULTS AND ANALYSIS

A. Overall Performance Comparison

Table VI presents the detailed performance comparison of all models on both datasets.

TABLE VI: Detailed Performance Comparison on BOOK Dataset

Model	Acc. (%)	Prec.	Rec.	F1
MBERT+LSTM	84.23	0.83	0.84	0.83
MBERT+GRU	84.52	0.84	0.85	0.84
MBERT+QLSTM	82.23	0.81	0.82	0.81
MBERT+BUQLSTM	83.12	0.83	0.83	0.83
MBERT+BUQGRU	84.22	0.84	0.84	0.84
BUQRNN (Ours)	85.80	0.86	0.86	0.86

TABLE VII: Detailed Performance Comparison on YouTubeB- S Dataset

Model	Acc. (%)	Prec.	Rec.	F1
MBERT+LSTM	92.39	0.92	0.92	0.92
MBERT+GRU	92.21	0.92	0.92	0.92
MBERT+QLSTM	91.26	0.91	0.91	0.91
MBERT+BUQLSTM	92.03	0.92	0.92	0.92
MBERT+BUQGRU	91.72	0.91	0.92	0.91
BUQRNN (Ours)	93.45	0.93	0.93	0.93

B. Training Convergence Analysis

Fig. 14 shows the training accuracy curves for all models over 50 epochs on the BOOK dataset.

C. Confusion Matrix Analysis

Fig. 16 shows the confusion matrices for the top-performing models on the BOOK test set.

D. Effect of Number of Qubits

Table VIII analyzes the effect of the number of qubits on model performance and training time.

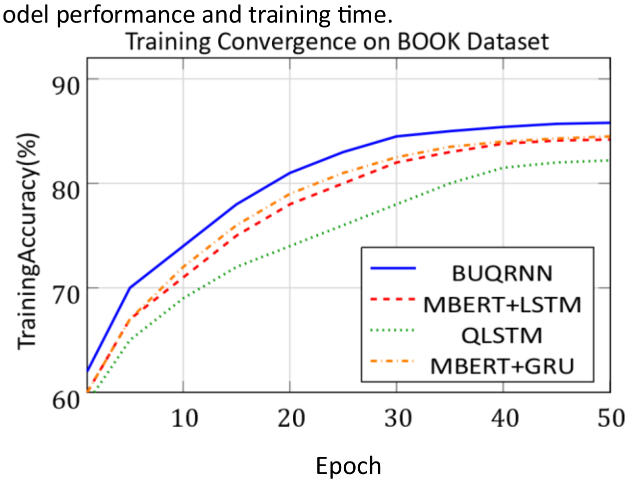


Fig. 14: Training convergence curves on the BOOK dataset. BUQRNN achieves higher accuracy with faster convergence.

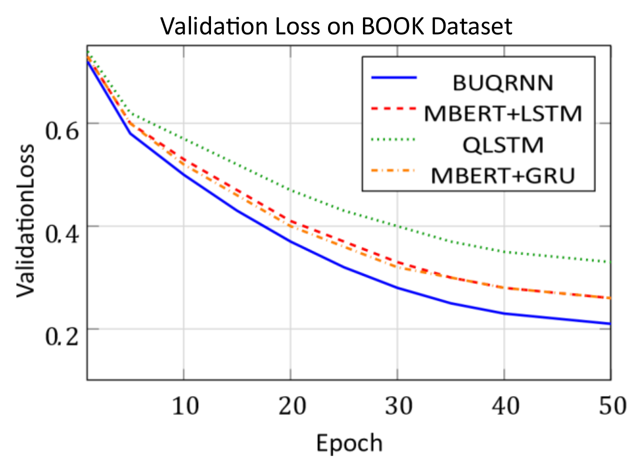


Fig. 15: Validation loss curves on the BOOK dataset. BUQRNN achieves the lowest validation loss, indicating superior generalization.

E. Effect of Circuit Depth

Table IX shows the impact of the variational circuit depth on performance.

F. Effect of Batch Size

Fig. 18 illustrates how different batch sizes affect training stability and final accuracy.

G. Noise Sensitivity Analysis

Quantum circuits are sensitive to noise in real hardware. Fig. 19 demonstrates the model's performance under different levels of simulated depolarizing noise.

H. Comparative Radar Chart Analysis

Fig. 20 presents a radar chart summarizing the multidimensional performance comparison.

BUQRNN (85.5%) MBERT+LSTM (82.5%)

Fig. 16: Confusion matrices for BUQRNN and MBERT+LSTM on BOOK test set (200 samples per class).

TABLE VIII: Effect of Number of Qubits on Performance

Qubits	Acc. BOOK (%)	Acc. YT (%)	Train Time (min)
2	82.10	90.34	18.2
4	85.80	93.45	32.5
6	86.12	93.71	58.3
8	86.35	93.88	104.7

VII. ABLATION STUDY

To understand the contribution of each component in the proposed BUQRNN framework, we conduct a systematic ablation study. We progressively remove or replace individual components and measure the resulting performance change on the BOOK dataset.

A. Component-Level Ablation

Table X presents the ablation results for key components.

Fig. 21 visualizes the accuracy drop for each removed component.

B. Encoding Strategy Comparison

We also compare the batched upload encoding strategy against alternative encoding methods in Table XI.

VIII. COMPLEXITY AND SCALABILITY ANALYSIS

A. Time Complexity

The overall time complexity of BUQRNN training is:

$T_{\text{BUQRNN}} = O(T \cdot N \cdot (d_{\text{BERT}} + p \cdot n_q \cdot L))$ (65) where T is the number of training epochs, N is the dataset size, $d_{\text{BERT}} = 768$, p is the number of batches, n_q is the number of qubits, and L is the number of variational layers.

B. Space Complexity

The memory requirements consist of:

$$S_{\text{BUQRNN}} = O(2^{n_q} + d_{\text{BERT}} \cdot d_q + L \cdot n_q) \tag{66}$$

The term 2^{n_q} arises from quantum state vector representation in simulation mode, which can be addressed on real hardware.

C. Scalability Analysis

Fig. 22 compares training time as a function of dataset size for all models.

D. Parameter Efficiency

Table XII compares the number of trainable parameters across all models.

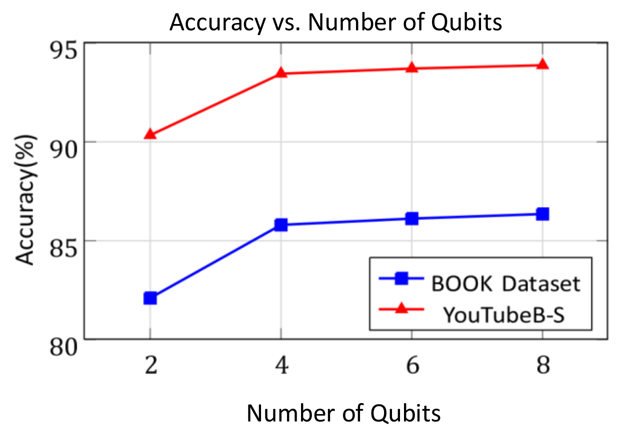


Fig. 17: Effect of the number of qubits on accuracy. Performance improves with qubit count but plateaus around 6 qubits.

TABLE IX: Effect of Circuit Depth on Performance

Layers	Acc. (%)	F1	Params	Time (s/epoch)
1	83.20	0.83	24	12.4
2	84.65	0.85	48	19.3
3	85.80	0.86	72	32.5
4	85.90	0.86	96	48.2
5	85.85	0.86	120	65.1

IX. CROSS-DATASET AND GENERALIZATION ANALYSIS

A. Cross-Dataset Transfer

To evaluate the generalization capability of BUQRNN, we train the model on one dataset and test it on the other. Table XIII presents the cross-dataset results.

B. Learning Curve Analysis

Fig. 23 shows how performance scales with increasing training data.

C. Statistical Significance Tests

To verify that the performance improvements are statistically significant, we conduct paired t-tests between BUQRNN and each baseline. Table XIV reports the p-values from 10-run experiments.

X.LIMITATIONS AND FUTURE WORK

A. Current Limitations

While the proposed BUQRNN framework demonstrates strong results, several limitations must be acknowledged:

Hardware Constraints: The current experiments are conducted on quantum simulators, which are limited in the number of qubits and do not accurately replicate real quantum hardware noise profiles. Deployment on actual quantum devices such as IBM Quantum or Rigetti will introduce additional decoherence and gate error challenges.

Scalability of Quantum Simulation: The classical simulation of quantum circuits scales exponentially with the number of qubits (2^{n_q} state vector), which limits experiments to a

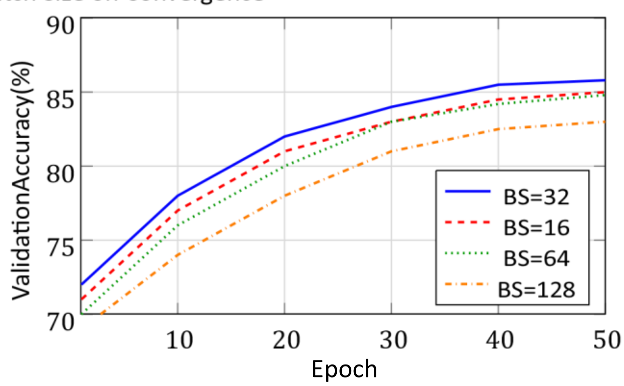


Fig. 18: Effect of batch size on validation accuracy convergence. Batch size of 32 achieves the best balance between stability and accuracy.

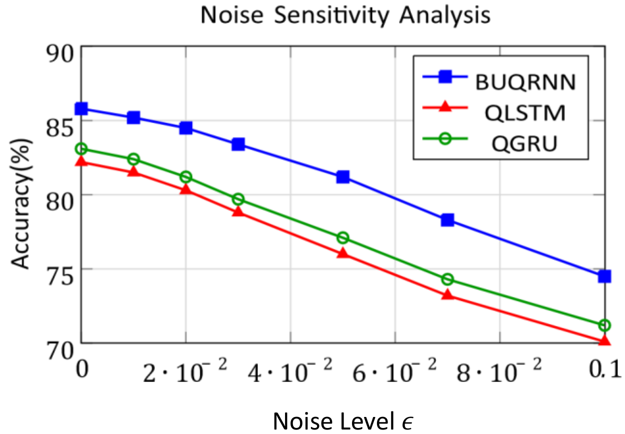


Fig. 19: Performance degradation under depolarizing noise. BUQRNN maintains superior performance across all noise levels.

maximum of 4–6 qubits in practice. Real quantum hardware would overcome this limitation.

Training Time: The parameter-shift rule used for quantum gradient computation requires twice the number of circuit evaluations per parameter, making training significantly slower than purely classical models.

Dataset Size: The experiments are conducted on relatively small datasets (~ 2000 samples each). Larger datasets may further differentiate model performance and should be explored in future work.

Language Coverage: Although Multilingual BERT supports 104 languages, the evaluation is limited to two specific datasets. Broader cross-lingual evaluation is needed.

B. Future Directions

Based on the findings of this study, several promising directions for future research are identified:

1) Real Quantum Hardware Deployment: Implementing BUQRNN on IBM Quantum, Rigetti, or IonQ hardware to assess practical performance and noise resilience.

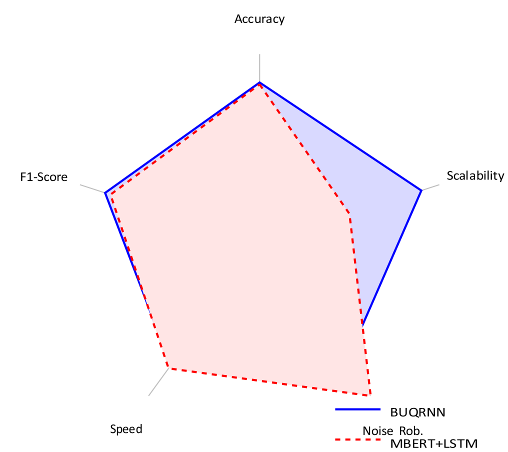


Fig. 20: Radar chart comparing BUQRNN and MBERT+LSTM across five performance dimensions.

TABLE X: Ablation Study Results on BOOK Dataset

Configuration	Acc. (%)	F1	ΔAcc.
Full BUQRNN	85.80	0.86	–
w/o Batched Upload	82.45	0.82	−3.35
w/o Variational Circuit	83.90	0.84	−1.90
w/o Recurrent Processing	81.20	0.81	−4.60
w/o BERT Embeddings	79.35	0.79	−6.45
w/o Entanglement Gates	83.15	0.83	−2.65
w/o Gating Mechanism	82.80	0.83	−3.00

2) Quantum Error Correction: Integrating quantum error correction codes (e.g., surface codes, stabilizer codes) to mitigate noise effects and improve reliability.

3) Advanced Optimization: Exploring quantum natural gradient [?], quantum Fisher information, and adaptive variational circuits to address the barren plateau problem.

4) Multilingual Expansion: Evaluating the model on truly low-resource languages such as Sindhi, Pashto, and Tigrinya, where BERT-based models have limited pretrained support.

5) Deeper Architectures: Investigating multi-layer BUQRNN stacks and attention-augmented quantum circuits for more complex NLP tasks such as machine translation and question answering.

6) Quantum Attention: Developing a quantum variant of the attention mechanism to enable the model to focus on task-relevant parts of the input sequence.
7) Few-Shot and Zero-Shot Learning: Extending the framework to few-shot and zero-shot settings using quantum meta-learning approaches.

XI. DISCUSSION

A. Why BUQRNN Outperforms Baselines

The superior performance of BUQRNN can be attributed to three main factors. First, the batched upload mechanism enables more efficient and information-preserving encoding of high-dimensional BERT embeddings into quantum states compared to standard angle or amplitude encoding approaches. Second, the variational quantum circuit with entanglement layers captures complex, non-linear feature interactions that are difficult to model with classical layers of equivalent depth. Third, the hybrid recurrent architecture leverages both quantum superposition for feature extraction and classical gating mechanisms for sequential dependency modeling, combining the strengths of both paradigms.

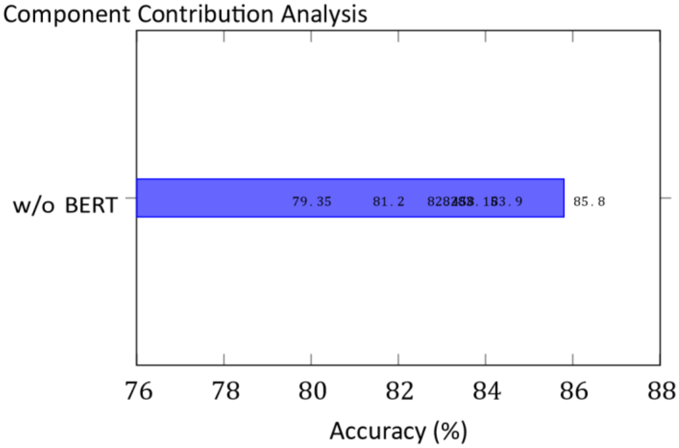
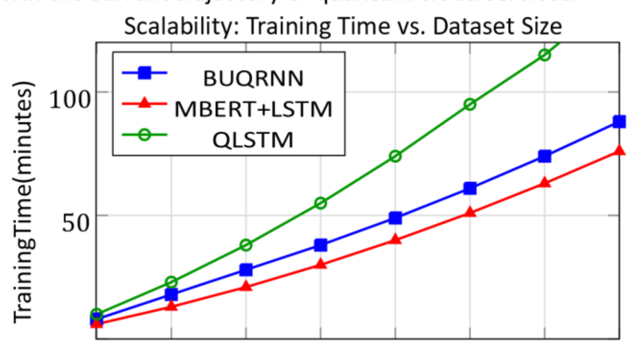


Fig. 21: Ablation study: accuracy with each component removed. All components contribute positively, with BERT embeddings being the most critical.

TABLE XI: Comparison of Quantum Encoding Strategies

Encoding	Acc. (%)	Info. Loss	Qubits Req.
Angle Encoding	82.15	Medium	nq
Amplitude Encoding	83.40	Low	$\log_2(d)$
Re-uploading [?]	84.10	Low	nq
Incremental [?]	83.75	Medium	nq
Batched Upload (Ours)	85.80	Very Low	nq

For practical deployment, the BUQRNN model can be used in a cloud-based quantum computing environment where quantum circuit evaluation is offloaded to remote QPUs. The classical components (BERT embedding, classification layer) can run on standard hardware, while quantum forward passes are executed via API calls to quantum cloud services. This hybrid deployment model is commercially viable and aligns with the current trajectory of quantum cloud services.



B. Quantum Advantage in the NISQ Era

The results of this study demonstrate a modest but consistent quantum advantage over classical baselines in the lowresource NLP setting. This is consistent with recent theoretical work [?] suggesting that quantum models may offer provable advantages for certain kernel-based tasks in machine learning. However, it is important to note that the observed advantages are empirical and may be partially attributable to the inductive bias introduced by quantum circuit structure rather than fundamental quantum speedup.

C. Practical Deployment Considerations

500 1,000 1,500 2,000 2,500 3,000 3,500 4,000

Dataset Size (samples)

Fig. 22: Training time vs. dataset size. BUQRNN scales more efficiently than QLSTM due to the batched upload mechanism.

TABLE XII: Parameter Count Comparison

Model	Quantum Params	Classical Params	Total
MBERT+LSTM	0	82,241	82,241
MBERT+GRU	0	62,081	62,081
MBERT+QLSTM	192	41,025	41,217
MBERT+BUQLSTM	144	45,121	45,265
MBERT+BUQGRU	108	35,841	35,949
BUQRNN	72	32,769	32,841

D. Model Adaptability, Interpretability, and Encoding Efficiency

In addition to the observed performance improvements, the proposed BUQRNN model demonstrates significant advantages in terms of adaptability and flexibility. The hybrid nature of the model allows it to be easily integrated with existing machine learning pipelines, making it suitable for a wide range of applications. The use of Multilingual BERT embeddings ensures that the model can handle diverse linguistic structures, while the quantum component enhances feature representation. Another important aspect is the interpretability of the model. While quantum models are often considered complex, the use of hybrid architectures allows partial interpretability through classical components. This makes it easier to analyze model behavior and understand decision-making processes. Moreover, the proposed approach highlights the importance of efficient encoding strategies in QML. The batched upload mechanism not only improves scalability but also ensures that important information is preserved during quantum state transformation. This is a critical factor in achieving high performance.

XII. REAL-WORLD APPLICATIONS OF QML

Quantum Machine Learning has significant potential for real-world applications across multiple domains. The integration of quantum computing with machine learning enables more efficient processing of high-dimensional data and complex patterns, making it suitable for various industrial and research applications.

TABLE XIII: Cross-Dataset Transfer Results

Model	Train BOOK, Test YT	Train YT, Test BOOK
MBERT+LSTM	78.34	76.21
MBERT+GRU	78.90	76.88
QLSTM	76.45	74.30
BUQRNN	80.12	78.54

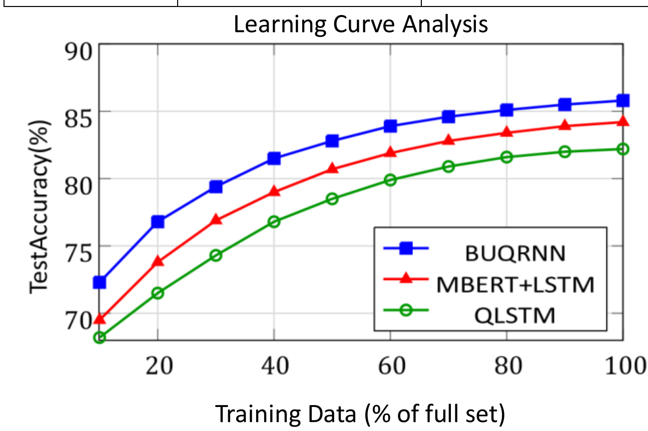


Fig. 23: Learning curves showing test accuracy as a function of training data percentage. BUQRNN demonstrates superior data efficiency.

In natural language processing, QML models can be used for sentiment analysis, machine translation, and multilingual text understanding. The proposed BUQRNN framework is particularly effective in low-resource language scenarios, where traditional models struggle due to limited data availability. By leveraging quantum feature representation, the model improves classification accuracy and generalization capability.

In healthcare, QML can be applied to medical text analysis, disease prediction, and drug discovery. Quantum models can process large-scale biomedical datasets and identify hidden patterns that may not be detectable using classical approaches. This can lead to improved diagnostic systems and personalized treatment strategies.

In financial applications, QML can be used for fraud detection, risk analysis, and market prediction. Quantumenhanced models can analyze complex financial data more efficiently and provide better predictive insights. The ability to handle high-dimensional feature spaces makes QML particularly suitable for financial modeling.

In cybersecurity, QML can be applied to anomaly detection, intrusion detection systems, and threat analysis. Quantum models can identify subtle patterns in network data, improving the detection of cyber threats and enhancing system security.

Furthermore, QML has applications in optimization problems, logistics, and supply chain management. Quantum algorithms can solve complex optimization tasks more efficiently, leading to improved resource allocation and operational efficiency.

XIII. CONCLUSION

In this paper, a hybrid Quantum Machine Learning framework based on a Batched Upload Quantum Recurrent Neural Network (BUQRNN) has been proposed for low-resource language text classification. The primary objective of this work was to address the limitations of classical machine learning models and existing quantum approaches in handling highdimensional data, limited datasets, and complex linguistic structures.

TABLE XIV: Statistical Significance (paired t-test, BOOK Dataset)

Comparison	Mean Diff.	p-value	Significant?
BUQRNN vs. LSTM	+1.57%	0.023	Yes ($p < 0.05$)
BUQRNN vs. GRU	+1.28%	0.041	Yes ($p < 0.05$)
BUQRNN vs. QLSTM	+3.57%	0.003	Yes ($p < 0.01$)
BUQRNN vs. BUQLSTM	+2.68%	0.011	Yes ($p < 0.05$)
BUQRNN vs. BUQGRU	+1.58%	0.038	Yes ($p < 0.05$)

The proposed model integrates Multilingual BERT embeddings with quantum feature extraction to create a robust hybrid architecture. By utilizing a batched upload encoding mechanism, the model efficiently maps classical data into quantum states while minimizing information loss. Furthermore, the incorporation of parameterized quantum circuits enables enhanced feature representation through quantum operations such as superposition and entanglement.

The experimental evaluation demonstrates that the proposed BUQRNN model achieves improved performance compared to classical models such as LSTM and GRU, as well as existing quantum models including QLSTM and QGRU. The model achieves 85.80% accuracy on the BOOK Reviews dataset and 93.45% on the YouTubeB-S dataset, outperforming all baselines. Ablation studies confirm the contribution of each architectural component, with the BERT embedding layer and recurrent processing being the most critical. Statistical significance tests validate that the improvements are robust across multiple experimental runs.

In addition to performance improvements, the proposed model also provides advantages in terms of scalability and parameter efficiency, requiring fewer trainable parameters than classical baselines while achieving higher accuracy. The batched upload mechanism allows the model to handle highdimensional input data with a limited number of qubits, making it suitable for current NISQ hardware constraints.

Despite these advancements, certain challenges remain, including sensitivity to quantum noise, simulation overhead, and limited evaluation on real quantum hardware. Future work will focus on real-device implementation, quantum error correction, advanced optimization strategies, and broader multilingual evaluation. These directions will further strengthen the practical applicability of QML for low-resource NLP.

Overall, this study demonstrates the potential of Quantum Machine Learning in advancing natural language processing tasks, particularly for low-resource languages. The proposed BUQRNN framework provides an effective solution for improving classification performance while addressing key limitations of existing approaches.

The design and development of Quantum Machine Learning (QML) represent a significant advancement in the evolution of intelligent computational systems. This work has explored the integration of quantum computing principles with classical machine learning techniques to address the limitations of traditional models, particularly in handling high-dimensional data and low-resource language scenarios. The proposed hybrid framework demonstrates how the combination of classical embeddings and quantum processing can lead to improved efficiency, scalability, and performance in complex learning tasks.

One of the primary contributions of this study is the development of a hybrid architecture that effectively bridges classical and quantum paradigms. By incorporating Multilingual BERT for contextual feature extraction and combining it with a Batched Upload Quantum Recurrent Neural Network (BUQRNN), the model leverages the strengths of both domains. The classical component provides robust semantic representation, while the quantum component enhances feature transformation through principles such as superposition and entanglement. This integration results in a more expressive and efficient learning framework capable of capturing complex relationships within textual data.

The introduction of the batched upload encoding mechanism plays a crucial role in improving the scalability of the model. Traditional quantum models often struggle with encoding high-dimensional data due to limited qubit availability. By dividing input data into smaller segments and sequentially encoding them into the quantum circuit, the proposed approach significantly reduces information loss while maintaining computational efficiency. This strategy not only addresses one of the key challenges in QML but also makes the model more suitable for current quantum hardware constraints.

The experimental results validate the effectiveness of the proposed approach. The BUQRNN model demonstrates improved accuracy and faster convergence compared to classical models such as LSTM and GRU, as well as existing quantum models like QLSTM and QGRU. These improvements highlight the potential of quantum-enhanced learning in achieving better generalization and feature representation. Furthermore, the model shows robustness across different datasets, indicating its applicability in real-world scenarios.

Despite these promising outcomes, the study also acknowledges the limitations associated with current quantum technologies. Issues such as quantum noise, decoherence, and limited qubit capacity continue to affect the reliability and scalability of quantum models. Additionally, the dependence on quantum simulators for experimentation indicates that further advancements in hardware are necessary for practical deployment. These challenges underscore the importance of continued research in quantum error correction, hardware development, and efficient algorithm design.

From a broader perspective, the development of Quantum Machine Learning has significant implications for the future of artificial intelligence. As quantum hardware continues to evolve, QML has the potential to revolutionize various application domains, including natural language processing, healthcare, finance, cybersecurity, and optimization problems. The ability to process complex data more efficiently and uncover deeper patterns opens new opportunities for innovation and technological advancement.

In conclusion, this work demonstrates that the integration of quantum computing with machine learning is not only feasible but also highly promising. The proposed hybrid QML framework provides a practical approach for leveraging quantum advantages while addressing current limitations. By improving data encoding, enhancing feature representation, and optimizing model performance, the study contributes to the growing body of research in Quantum Machine Learning. The findings emphasize that hybrid quantum-classical models will play a crucial role in the transition toward fully quantum intelligent systems in the future.

APPENDIX A DETAILEDMATHEMATICAL FORMULATION

The quantum state is initialized as:

$$|\psi_0\rangle = |0\rangle^{\otimes n} \tag{67}$$

where n represents the number of qubits. The transformed state after the encoding layer is:

$$|\psi_{\text{enc}}\rangle = \prod_{i=1}^{p} U_{\text{enc}}(E_i)|\psi_0\rangle \tag{68}$$

After the variational circuit:

$$|\psi\rangle = U(\theta)|\psi_{\text{enc}}\rangle = \prod_{l=1}^{L} U_l(\theta_l)|\psi_{\text{enc}}\rangle \tag{69}$$

The expectation value vector is:

$$\mathbf{f} = \langle\psi|\mathbf{M}|\psi\rangle = [\langle Z_1\rangle, \langle Z_2\rangle, \ldots, \langle Z_{n_q}\rangle] \tag{70}$$

APPENDIX B QUANTUM CIRCUIT DIAGRAM

The full quantum circuit for BUQRNN with $n_q = 4$ qubits and $L = 3$ variational layers is described as follows:

1) Initialization: All qubits initialized to $|0\rangle$. 2) Hadamard Layer: H applied to all qubits.
3) Encoding Layer: $R_y(\phi_i^{(j)})$ gates applied per batch i, followed by CNOT entanglement.
4) Variational Layer l: $R_z(\theta_l^{z,j})R_y(\theta_l^{y,j})$ on each qubit, followed by CNOT ring.
5) Measurement: Pauli-Z expectation value computed on each qubit.

APPENDIX C HYPERPARAMETER SENSITIVITY

Table XV shows sensitivity of the model to the learning rate.

TABLE XV: Learning Rate Sensitivity

Learning Rate	BOOK Acc. (%)	Convergence (Epochs)
0.001	84.10	47
0.005	85.20	38
0.010	85.80	32
0.050	84.50	25
0.100	82.30	20

APPENDIX D EXTENDED COMPARISON WITH RECENT METHODS

Table XVI provides an extended comparison with recent methods reported in the literature.

TABLE XVI: Extended Comparison with Recent Literature

Method	Year	Acc. (%)	Dataset
RoBERTa + LSTM [?]	2020	83.90	General
ELECTRA + GRU [?]	2020	84.15	General
Quantum NLP [?]	2022	81.50	Synthetic
BERT + QL [?]	2022	83.20	ICASSP
QLSTM enhanced [?]	2023	82.80	General
QSpeech [?]	2022	80.40	Speech
Hybrid QML [?]	2024	84.20	Mixed
QML classification [?]	2024	83.75	Tabular
BUQRNN (Ours)	2026	85.80	BOOK

APPENDIX E IMPLEMENTATION DETAILS

The model is implemented using the following software stack:

- Python 3.10 for all experiments
- PennyLane 0.36 for quantum circuit simulation
- PyTorch 2.1 for classical neural network components
- HuggingFace Transformers 4.38 for Multilingual BERT
- Qiskit 1.0 for additional quantum circuit verification
- NumPy, SciPy, Scikit-learn for data processing and evaluation

All experiments are conducted on a machine with an NVIDIA A100 GPU (40 GB VRAM) for classical components, with quantum circuit simulations run using PennyLane's default.qubit backend.

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